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SECJ2013 - DSA

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QUIZ 2

ALGORITHM EFFICIENCY

Determine the big notation of the following expressions:

A. $2n + 5n^2 \Rightarrow O(n^2)$

B. $3n \log_2 n + 5n - 10n + 5n \Rightarrow O(n \log_2 n)$

C. $3 + \log_2 n + 2n + n^2 + 2^n \Rightarrow O(2^n)$

D. $1 + 2 + 3 \Rightarrow O(1)$

E. $1n - 2n + 1n \Rightarrow O(1)$

List the order of the above expression from the most simple to the complex one:

$$D < E < B < A < C$$

ALGORITHM EFFICIENCY

Calculate the number of steps and determine the big-O notation of the following program:

Statement	Number of Steps
<code>int start = 0, stop = 10;</code>	1
<code>for (int i = start; i <= stop; i++) {</code>	$n - 0 + 1 = n + 1$
<code> cout << "Hello\n";</code>	$(n+1) \cdot 1 = (n+1)$
<code>}</code>	0
Total steps	$1 + (n+1) + (n+1) + 0 \Rightarrow 2n+3$
Big-O notation	O(n)

ALGORITHM EFFICIENCY

Calculate the number of steps and determine the big-O notation of the following program:

Statement	Number of Steps
<code>int i, j, n = 8;</code>	1
<code>for (i = 1; i < n; i++) {</code>	$n - 1$
<code> j = 1;</code>	$(n - 1) \cdot 1 = n - 1$
<code> while (j <= i) {</code>	$n(n - 1) / 2$
<code> j++;</code>	$n(n - 1) / 2$
<code> cout << i + j << "\n";</code>	$n(n - 1) / 2$
<code> }</code>	0
<code>}</code>	0
Total steps	$\begin{aligned} &1 + 2(n - 1) + 3[n(n - 1) / 2] \\ &= 1 + 2n - 2 + 3[(n^2 - n) / 2] \\ &= 2n - 1 + (3/2)n^2 - (3/2)n \\ &= (3/2)n^2 + (1/2)n - 1 \end{aligned}$
Big-O notation	$O(n^2)$

SIMPLE SORT

1. Given that `num` array has been defined as follows:

```
int num = [ 2, 1, 4, 3, 5];
```

List out the content of the `num` array after passes 1 and 2 if it was sorted by using bubble, selection, and insertion sort in ascending order.

Passes	Bubble Sort	Selection Sort	Insertion Sort
After Pass 1	[1, 2, 3, 4, 5]	[2, 1, 4, 3, 5]	[1, 2, 4, 3, 5]
After Pass 2	[1, 2, 3, 4, 5]	[2, 1, 3, 4, 5]	[1, 2, 4, 3, 5]

2. For an array `[4, 3, 1, 2]`, determine the type of sorting (ascending/descending) and the implemented algorithm (bubble/improved bubble/selection/insertion) for the following output:

```
Pass1: 4 3 2 1
```

```
Pass2: 4 3 2 1
```

```
Num. of comparisons: 5
```

```
Num. of swaps: 1
```

Answer: Descending order and Improved Bubble sort. It is because the array go through 2 passes only. The subsequent passes is not continued since the array is already sorted.

ADVANCE SORT

Given two halves of array to be merged in the last merge operation as follows:

[1, 7, 9,] [2, 4, 6, 10]

Complete the tracing of the value of first1, first2, value comparison, and the temporary array content until all items are fully copied and sorted into the temporary array:

first1-	first2	value comparison	tempArray
0	3	1 < 2	[1, null, null, null, null, null, null]
1	3	2 < 7	[1, 2 , null, null, null, null, null]
1	4	4 < 7	[1, 2 , 4 , null, null, null, null]
1	5	6 < 7	[1, 2 , 4 , 6 , null, null, null]
1	6	7 < 10	[1, 2 , 4 , 6 , 7 , null, null]
2	6	9 < 10	[1, 2 , 4 , 6 , 7 , 9 , null]
-	6	-	[1, 2 , 4 , 6 , 7 , 9 , 10]

ADVANCE SORT

For an array `int list = [12, 6, 16, 4, 14, 2]`, with pivot value is 12 (the first element), trace the position of the bottom (**B**) and top (**T**) index and list the content of the array after each 1st, 2nd and 3rd swapping operation in the `partition(list, 0, 5)` function of a quick sort algorithm (ascending order).

Initial position => [12, 6, 16, 4, 14, 2]
 B **T**

After 1st swap => [2, 6, 16, 4, 14, 12]
 B **T**

After 2nd swap => [2, 6, 12, 4, 14, 16]
 B **T**

After 3rd swap => [2, 6, 4, 12, 14, 16]
 BT

What is the cut point returned by the `partition(list, 0, 5)` function and the next two calls of quick sort function: `quickSort(list, ?, ?)`?

cut point => 3
`partition(list, 0, 3)`
`partition(list, 4, 5)`